

Jonathan Setiawan

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SUMMARY / PROFESSIONAL OBJECTIVE

Robotics and AI researcher with interest in practical application of autonomous systems within high-impact sectors. Proven expertise in engineering robust perception-to-action pipelines and software for real-world deployment.

EDUCATION

Ritsumeikan University (立命館大学) **Osaka, Japan**
College of Information Systems Science and Engineering *Graduation Date: April 2028*
Awards: Murai SIIX Scholarship (2025 & 2026), JASSO Monbukagakusho Honors Scholarship (2024)
Certifications: IELTS 8.0 (2022) *CGPA: 4.35/5.0*
Relevant Coursework: Network systems, Artificial Intelligence, Project Based Learning, C Programming, DSA

SKILLS

Python, C, C++, Java, JavaScript/TypeScript, MySQL, ROS2 (Humble), Gazebo, Networking, OpenCV, , PyTorch, TensorFlow, SLAM, Docker, Git, React, CMake, LaTeX, Linux (Ubuntu, Fedora, CachyOS), Go, PHP, Rust

RESEARCH & LEADERSHIP EXPERIENCE

FPTジャパン株式会社 **Tokyo, Japan**
Physical AI Team Associate AI Engineer *May 2026 - Present*

- Deployed and evaluated imitation-learning policies (ACT, modified Diffusion Policy, SmolVLA) using the LeRobot framework for physical AI applications. Trained and tested MolmoAct (Action Reasoning Model) to integrate 2.5D visual perception with grounded trajectory planning for imitation learning.

iCOM技研株式会社 **Hyogo, Japan**
R&D Systems Engineer Intern *Jul 2025 - Present*
Multi-robot AI Palletizing and Depalletizing System

- Architected a palletizing and depalletizing system for UR5e, UR10e, and UR20 robots using RANSAC-based surface modeling and temporal filtering, achieving ≥ 0.97 confidence score.
- Automated industrial hand-eye mapping using Tsai-Lenz ($AX=XB$) methodology and the RTDE API, reducing spatial error to $< 2\text{mm}$ for stable continuous sorting operations.

Ritsumeikan University (立命館大学) **Osaka, Japan**
RiOne (Robocup @Home League) Robotics Club Computer Vision Team Lead *Jul 2024 - Present*

- Architected a multi-node ROS 2 Humble perception stack and sensor-fused navigation on TurtleBot3 WafflePi with face re-identification and temporal tracking for persistent HRI and autonomous obstacle avoidance.
- Engineered a self-improving object recognition pipeline and achieved 3rd Place for Open Challenge and Best Poster Award in RoboCup Japan Open 2026 (Bridge Division).

PROJECT EXPERIENCE

SciBERT-CTFT: Scientific Abstract Classifier via Contrastive Learning Fine-tuning
Artificial Intelligence Course Project *Dec 2025 - Jan 2026*

- Fine-tuned a SciBERT model using the SetFit framework to classify scientific literature with 99.61% accuracy, implementing a novel context injection data augmentation technique using TF-IDF maps.
- Validated model robustness via 5-fold cross-validation, demonstrating consistent performance (> 0.99 F1-score) across domains like Bioinformatics and Neuroscience despite limited labeled data.

EFT-VPR: Event Forecasting Transformer for Visual Place Recognition
Personal Project *Feb 2026 - Mar 2026*

- Engineered an event-based localization stack utilizing a 3-layer SNN encoder and an 8-head autoregressive Transformer with InfoNCE contrastive learning, improving Recall@1 by 13.8% in a 65,992-frame dataset.
- Designed a production-ready VPR pipeline leveraging FAISS (IndexFlatIP/GPU) for top-K cosine retrieval and dropout-robust forecasting, reducing median localization error by 27.8% (81.9m to 59.1m).

Foveated Teleoperation Task-based Study
PBL5 project *Apr 2026 - Present*

- Architected a real-time foveated teleoperation pipeline streaming headset eye-tracking data via TCP/UDP to an encoding server, and onward to a rover mini-PC with ESP32 motor control.
- Engineered a 90 Hz telemetry logger and implemented gaze-signal filtering with velocity-based fixation classification for low-latency robotic control.